

Mathematical Communication: Remarks from Tao on Writing Papers

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This document is based on Terence Tao’s writing advice from his blog. It has been reorganized and shortened. It was presented on March 22 and 29, 2024. There are exceptions to every rule, but here are some rules of thumb.

1 The introduction

Sell the key points.

1. Get to the point, quickly. Attention is limited.
2. State main results in appropriate detail. Consider writing a special case of your main theorem before stating it in full generality.
3. Explain new aspects, merits, consequences of your main results by comparison to existing work in the literature.

Yuval adds that some explanations can be deferred to a “Related Results” section. When citing other papers, write all the authors’ names, not just something like “[4]”. Also, write the year, always. It helps advanced and new researchers form a coherent story related to the mathematical content. Andrew agrees that the year information should be included, perhaps in the format of the citation mark (e.g., “[F77]” for a paper published in 1977). This can be done automatically in LaTeX with the correct settings.

4. Can one drop hypotheses in the main results and still have true statements? Are there relevant counterexamples showing the “sharpness” of your results? Give these.

2 Organization

The reader only knows the structure of your argument if you tell them.

1. Outside of the introduction, readers expect to see Definitions, followed by Lemmas, followed by Theorems, but they do not know the relationship of

each result/section to the overall point of the paper. The author has the “curse of knowledge”; in other words, the author worked with the content of the paper and presumably understands the relationships among the ideas of the paper. Therefore, the author may be unable to recognize when these ideas are not well explained. For readers, the relationship between various ideas in the paper may not be obvious, and hierarchical structure with plenty of signposting text can help fix this. See the next points.

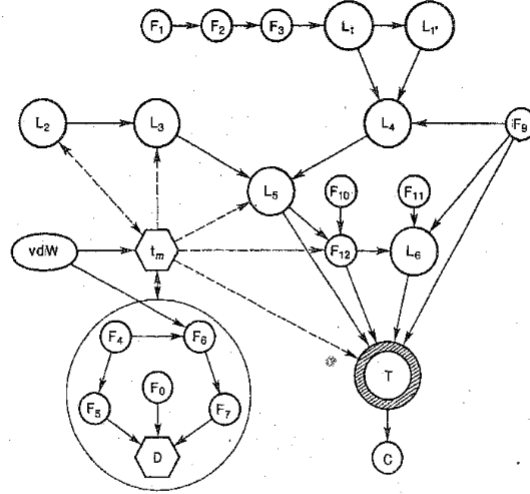
2. Sections of the paper should be major milestones for your results.

Andrew suggests the following test of the worth of a section’s name—call it the “Step 2 test”: If a title like “Step 2” or “Step 3” is the **only** possible name for a section, then you should reorganize the paper.

A section should be named something informative, not “Technical lemmas.” Some possible good titles include “Counting polynomial configurations” or “Proof of Theorem 1.1: Small arcs.”

Minimize the use of math symbols in titles of sections.

3. Explain the major milestones in the introduction, perhaps in a little subsection called “Roadmap,” “Structure,” or “Outline.”
4. Create a map explaining the implications and logical dependencies of the results of the paper. Often there will be a handful of key results that themselves depend on several less important results, and this structure should be obvious from the map. Whether you include this map in the paper is a separate decision, but the process of creating the map can clarify the milestones of the paper. One reason not to include this map is if the logical structure is “too” linear, such as “Theorem A implies Theorem B implies Theorem C.” Here is an example of such a map that is worth inclusion, from the article “On sets of integers containing no k elements in arithmetic progression” published in 1975 by E. Szemerédi:



The diagram represents an approximate flow chart for the accompanying proof of Szemerédi's theorem. The various symbols have the following meanings: F_k = Fact k , L_k = Lemma k , T = Theorem, C = Corollary, D = Definitions of B, S, P, α, β , etc., t_m = Definition of t_m , vdW = van der Waerden's theorem, F_0 = "If $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is subadditive then $\lim_{n \rightarrow \infty} \frac{f(n)}{n}$ exists".

Aggressively hide less important content.

1. Side remarks, tedious computations, and overly technical proofs should be delayed as much as possible. Their inclusion too early in the paper wastes the attention of the reader on irrelevant information and increases the chance that the reader will stop reading.

Yuval adds: Don't use appendices; instead, use normal sections, just placed later. Don't look down on your own work by calling it "technical." Often the technical part is precisely where the innovation is, in which case, care should be taken to appropriately sell the audience on the importance of the work.

2. If possible, each section should be summarizable in one result. That is, ask yourself whether the reader can safely forget everything except one result from the section. This allows your paper to more easily penetrate the mind of the average reader, who will forget much of what they read.
3. If you state a lemma, it should be psychologically important or used more than once. For example, a key subargument that is only used once probably qualifies as a lemma. Any technical lemma that is used in several places probably qualifies as a lemma. However, often there are subarguments that are less important than other ones and only appear once; Tao

recommends folding these into the proofs of the lemmas that are actually important.

Yuval disagrees with this last suggestion and argues for a modular structure, in which each step of an argument is given its own lemma, thus allowing a reader to check part of an argument, forget about it, and then return later to check another part, and so on.

Some comments regarding the appropriate level of detail.

1. Do not rewrite proofs of standard lemmas without a good reason.

A particular result may be new to you, but well known. Including a proof of such a result without a good reason to do so shows that you are not familiar with the community to which you are writing.

Yuval adds: If you can cite a result from a textbook, do that. If that source is hard to access or hard to read because of outdated language, then one can consider rewriting it in modern language (along with adding an appropriate comment to this effect).

2. Give precise references to cited works. For example, instead of referring to an entire book, refer to Proposition 7.1 in the book instead; same for papers.

Search for a good reference for “well known” results. You can ask people in the field for this if you really cannot find it.

3. Statements of lemmas and theorems should be **self-contained**.

Self-contained means that all of the terms in the result are defined in the paper, and all additional assumptions are stated in the text of the theorem or lemma.

A common example of what to avoid: “In Section 2, all functions are assumed to be nondecreasing and continuously differentiable.” Readers mostly do not read linearly. If this sentence is at the beginning of the paper or at the beginning of a section, it will be very easy to miss if lemmas do not alert the reader to the extra assumptions. If there are assumptions that are both very long and repeatedly needed, then one could instead write something like the following:

In this section, all of our functions f satisfy the following assumption:

f is nondecreasing, continuously differentiable, and satisfies
some other condition that takes several lines of text to write out. (1)

Later, the statement of a lemma may read:

Let f satisfy Assumption (1). Assume also that f is green.
Then...

Another option with assumptions that are very long but only used two or three times is to state one lemma with the assumptions written out inside the lemma, then write subsequent lemmas like so:

Let f be as in Lemma 3.1. Then...

Again, while it is debatable whether each of these options is entirely “self-contained,” the point is to **be clear, even when the reader has not read from the beginning.**